

Technology Stack

OptiCrop — Smart Agricultural Production Optimization Engine

Languages & Core Libraries

- Python 3.x — primary development language
- NumPy — numerical computing and array operations
- Pandas — data manipulation, cleaning, and analysis

Machine Learning

- Scikit-learn — model training (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means), scaling, train/test splitting, evaluation metrics
- SciPy — supporting statistical/scientific computation

Visualization

- Matplotlib — base plotting (distributions, elbow curve, confusion matrix)
- Seaborn — statistical visualization (heatmaps, boxplots, count plots, pairplots)

Web Framework

- Flask — lightweight WSGI web framework serving the application and the prediction endpoint
- HTML5 + Bootstrap 5 — front-end structure and styling

Development Environment

- VS Code — code editing, Flask app development, integrated terminal
- Jupyter Notebook (via VS Code's Jupyter extension) — exploratory data analysis and model training
- Python venv — isolated dependency management
- Git & GitHub — version control and project hosting